

Native Time–Energy Reciprocity

Replacing a borrowed justification (Doc. 306)

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Doc. 306 Native time–energy reciprocity — replacing a borrowed justification

Abstract. Documents 025 and 063 justify the absence of a cosmic initial singularity by Heisenberg's energy–time uncertainty relation. That justification is here *replaced*, not refuted: the conclusion (no singular beginning) stands, the derivation is replaced by a native one. Three defects are named — an inverted implication (a finite Δt yields a *finite* bound, not $\Delta E \rightarrow \infty$), a misapplication (the Δt of the relation is an observable's characteristic change time, not a cosmic age), and a breach of style ("irrefutably" is not a ledger term). The replacing chain is native: from $T \cdot m = 1$ (Doc. 003/010), hence $T \cdot E = 1$, it follows that $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$; but FFGFT time carries a minimal, non-vanishing increment $d_1 = 100 \xi_1 = 1/75$, fixed by $\xi = 4/30000$. T is therefore bounded below and E bounded above — an infinite density is structurally excluded. In addition $\partial T^4 = \emptyset$: the question of a "beginning" presupposes a boundary that a compact torus does not have.

.1 What is being replaced

Docs. 025 and 063 both contain the passage that Heisenberg's uncertainty relation $\Delta E \cdot \Delta t \geq \hbar/2 = 1/2$ (natural units) "irrefutably proves" a classical Big-Bang singularity to be impossible, and infer:

- a temporal beginning would mean Δt finite;
- this leads to $\Delta E \rightarrow \infty$;
- hence the universe must have existed eternally.

The *conclusion* — no singular beginning, instead a finite core with a minimal scale L_0 from ξ — is retained. The *derivation* is replaced. Three reasons.

.2 Three defects of the borrowed justification

(M1) Inverted implication. The inference " Δt finite $\Rightarrow \Delta E \rightarrow \infty$ " is invalid. From $\Delta E \geq 1/(2\Delta t)$, a *finite* Δt yields a *finite* lower bound on ΔE ; a beginning a finite time ago would give a tiny, harmless bound. $\Delta E \rightarrow \infty$ follows only from $\Delta t \rightarrow 0$, i.e. from the *instant* $t = 0$, not from a finite beginning. This is not a slip but a reversed implication.

(M2) Misapplication. Energy–time uncertainty is not a canonical uncertainty relation. By Pauli's theorem there is no self-adjoint time operator conjugate to a Hamiltonian bounded below; the rigorous form (Mandelstam–Tamm) defines $\Delta t = \Delta A/|d\langle A \rangle/dt|$ — the characteristic *change time of an observable*, not an elapsed duration. The "age of the universe" is not an admissible Δt for this relation.

(M3) Breach of style. "Irrefutably proves" is a strength of assertion that does not occur in the FFGFT ledger. The ledger knows *proven* (derived from ξ), *restricted*, and *blocked*; a statement is certified as sufficiently precise to be decidable, not as irrefutable.

The overarching point. Heisenberg's relation is not false within the QM framework, but it is *projected*: it belongs to the description level that FFGFT understands as a projection from the deeper structure. An argument against the cosmic singularity that leans on it does not leave the projection — it circulates within it. This is exactly what Doc. 298 warns of: one does not exit the projection by importing its theorems.

.3 The replacing native chain

(1) Exact reciprocity. The time–mass duality (Doc. 003, cf. Doc. 010) reads

$$T \cdot m = 1, \quad \text{and with } E = m \text{ (natural units)} \quad \boxed{T \cdot E = 1} \quad (.1)$$

This is an *equation*, not an inequality — a sharper statement than $\Delta E \cdot \Delta t \geq 1/2$, and a native one.

(2) Time has a minimal increment. FFGFT time is the cumulated defect of the unrolled winding,

$$D(k) = \sum_{n \leq k} 100 \xi_n \rightarrow \ln\left(1 + \frac{k}{75}\right), \quad \xi_{n+1} = \xi_n (1 - 100 \xi_n), \quad (.2)$$

with the first step

$$d_1 = 100 \xi_1 = 100 \cdot \frac{4}{30000} = \frac{1}{75}. \quad (.3)$$

To be precise: $D(k)$ grows without bound as $k \rightarrow \infty$ (logarithmically). The singularity-free statement is not " D never diverges" but: $D(k)$ is *finite at every finite k* , and there is no point at which D or its increments diverge. The first step is finite and fixed by ξ ; no time increment is smaller than d_1 .

(3) The inference. From $T \cdot E = 1$,

$$E \rightarrow \infty \iff T \rightarrow 0. \quad (.4)$$

But no state with $T = 0$ exists in FFGFT: time is bounded below by a finite increment fixed by ξ . Hence E is bounded above — an infinite energy density is *structurally* excluded, not by a statement about measurement precision. The exact numerical bound depends on the normalisation and is not asserted here as a number; the structural statement stands without it.

(4) The boundary question. The torus is compact and boundaryless, $\partial T^4 = \emptyset$ (Doc. 302; cf. Doc. 263, Doc. 296). The question “what was at $t = 0$?” presupposes a boundary at which a beginning could sit. A compact torus has none. The question is therefore not wrongly *answered* but wrongly *posed*: it imports the decompactified projection as ontology. What standard cosmology reads as a singular beginning is the point at which the *projection* ends — not the world.

.4 An observation on the genesis

It is notable that 025 and 063 already point to ξ in their *result*: the singularity is said to be “replaced by a tiny but finite core with minimal length scale L_0 (from ξ)”. The answer therefore already came from ξ ; only the ladder climbed to reach it was borrowed. Why the import was made although Doc. 003/010 was available is a *historical question about the genesis of the corpus*, not a systematic one. It is marked as such here and left unanswered; any answer would be speculation.

.5 Status

Correction / replacement. The conclusion (no singular beginning; a finite core with a minimal scale set by ξ) is unchanged. The justification in Docs. 025 and 063 is *no longer authoritative* for this point; it is not recorded as refuted, but as superseded by the native chain.

Proven (internal to FFGFT). $T \cdot m = 1$ (Doc. 003/010); $D(k)$ and $d_1 = 1/75$ from ξ and the recursion (Doc. 295); $\partial T^4 = \emptyset$ (Doc. 302). The inference $E \rightarrow \infty \iff T \rightarrow 0$ follows algebraically from the reciprocity.

Open. The exact numerical upper bound on E requires a fixed normalisation and is not asserted here. The connection $R_H \rightarrow \ell_1$ remains open (P20, cf. Doc. 267); the cosmological sector as a whole remains the least determined area of the corpus.